

The Second Connectivity Tax

A complete entropy certificate for $a_*(1) \geq 0.462$

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Abstract

For a finite nonempty tree T , let $\mu(T)$ be the mean size of a uniformly random independent set and let $\alpha(T)$ be its independence number. Put

$$a_*(1) = \inf_T \frac{\mu(T)}{\alpha(T)}.$$

We prove

$$a_*(1) \geq \frac{231}{500} = 0.462.$$

The proof starts from the exact entropy-partition decomposition and adds two connectivity taxes. The first is a collision term under a comparison hard-core measure. The second is exact Bethe total correlation after conditioning on the nonisolated part of $T[B]$, where B is the complement of a maximum independent set. A global rooting argument assigns every remaining degree-3 or degree-4 deficit exactly once. The only infinite local families are closed analytically by log-convexity and elementary tail bounds; all remaining scalar inequalities are enclosed by exact rational intervals for logarithms. The accompanying verifier uses only integer and rational arithmetic.

1 Statement and the adjusted master inequality

For a finite graph G , write

$$Z_G = |\mathcal{I}(G)|, \quad \mu(G) = \frac{1}{Z_G} \sum_{I \in \mathcal{I}(G)} |I|, \quad \alpha(G) = \max_{I \in \mathcal{I}(G)} |I|.$$

Let

$$c = \frac{\log(34/5)}{6 \log 2}, \quad \gamma = \frac{1}{2} - c, \quad r_0 = \frac{231}{500}, \quad a = \frac{1}{2} - r_0 = \frac{19}{500}.$$

Theorem 1 (Second connectivity tax). *Every finite nonempty tree T satisfies*

$$\mu(T) \geq \frac{231}{500} \alpha(T).$$

Consequently $a_(1) \geq 0.462$.*

Fix a maximum independent set A of T . Put

$$\alpha = |A|, \quad B = V(T) \setminus A, \quad K = T[B], \quad e = |E(K)|,$$

and for $v \in B$ put

$$d_v = |N(v) \cap A| \geq 1.$$

Since T is a tree and A is independent,

$$\sum_{v \in B} (d_v - 1) = \alpha - 1 - e. \quad (1)$$

For $d \geq 1$ define

$$\ell_d = \left\lceil \log \frac{5/4}{1 + 2^{-d}} \right\rceil_+, \quad \Delta_d = \gamma(d - 1) - \frac{\ell_d}{2 \log 2}.$$

The exact entropy-partition decomposition gives

$$\mu(T) - c\alpha \geq \gamma(1 + e) + \sum_{v \in B} \Delta_{d_v} + \frac{R_A}{2 \log 2} + \frac{D_{\text{KL}}(P \| Q_{1/4})}{2 \log 2}. \quad (2)$$

Here P is the law of $J = I \cap B$ for a uniform independent set I of T , $Q_{1/4}$ is the hard-core measure of activity $1/4$ on K , and R_A is the overlap remainder defined below. The omitted coordinate-comparison remainder is nonnegative.

Subtract $(r_0 - c)\alpha$ and use (1). Set

$$s_d = a(d - 1) - \frac{\ell_d}{2 \log 2}, \quad D_d = (-s_d)_+, \quad t_d = \frac{(s_d)_+}{d}. \quad (3)$$

Then

$$\mu(T) - r_0\alpha \geq a(1 + e) + \sum_{v \in B} s_{d_v} + \frac{R_A}{2 \log 2} + \frac{D_{\text{KL}}(P \| Q_{1/4})}{2 \log 2}. \quad (4)$$

Also

$$s_d = d t_d - D_d. \quad (5)$$

2 Exact overlap and conditional factor identities

Let $w_v = 2^{-d_v}$ and let Q_w be the hard-core measure on K with vertex activities w_v . For $J \in \mathcal{I}(K)$ define

$$O(J) = \sum_{v \in J} d_v - |N_A(J)| = \sum_{x \in A} (|J \cap N(x)| - 1)_+.$$

Counting first on B gives

$$Z_T = 2^\alpha \sum_{J \in \mathcal{I}(K)} 2^{-|N_A(J)|} = 2^\alpha Z_K(w) \mathbb{E}_{Q_w} 2^{O(J)},$$

so

$$R_A = \log \mathbb{E}_{Q_w} 2^{O(J)}. \quad (6)$$

Split

$$R = \{v \in B : \deg_K(v) > 0\}, \quad C = B \setminus R.$$

Every $v \in C$ is isolated in K . Therefore under Q_w the indicators $(\mathbf{1}_{v \in J})_{v \in C}$ are mutually independent and

$$p_d := Q_w(v \in J) = \frac{1}{2^d + 1} \quad (d = d_v). \quad (7)$$

For $x \in A$ put $X_x = |J \cap N(x) \cap C|$. Since $O(J) \geq \sum_x (X_x - 1)_+$ and the factors $2^{(X_x - 1)_+}$ are increasing functions of independent Bernoulli variables, Harris' inequality and (6) imply

$$\frac{R_A}{2 \log 2} \geq \sum_{x \in A} \mathcal{R}_x, \quad (8)$$

where, if the C -neighbour degrees of x are $d_1, \dots, d_q$,

$$\mathcal{R}(d_1, \dots, d_q) = \frac{1}{2 \log 2} \log \frac{\prod_i (1 + p_{d_i}) + \prod_i (1 - p_{d_i})}{2}. \quad (9)$$

Indeed, for X a sum of independent Bernoulli variables,

$$\mathbb{E} 2^{(X-1)_+} = \frac{1}{2} \left(\prod_i (1 + p_i) + \prod_i (1 - p_i) \right).$$

We next extract a second, independent budget from the KL term. Condition on the configuration on R . Since C is isolated in K , the conditional reference law under $Q_{1/4}$ is the product $\text{Ber}(1/5)^C$. Under P , the conditional law on C is proportional to

$$\prod_{x \in A} \psi_x(\xi_{N_C(x)}), \quad \psi_x = \begin{cases} 2, & \text{if no neighbour of } x \text{ in } B \text{ is occupied,} \\ 1, & \text{otherwise.} \end{cases} \quad (10)$$

After the R -configuration is fixed, each factor is either constant or has the displayed two-level form. Its factor graph is a subgraph of the original tree and hence a forest.

Lemma 2 (Exact conditional Bethe budget). *For every fixed R -configuration, let $P_{C|R}$ be the law in (10). Then*

$$\text{TC}(P_{C|R}) = \sum_{x \text{ active}} \left(\sum_{v \in N_C(x)} H(S_v) - H(S_{N_C(x)}) \right). \quad (11)$$

Moreover

$$D_{\text{KL}}(P \| Q_{1/4}) \geq \mathbb{E}_R \text{TC}(P_{C|R}). \quad (12)$$

Proof. For a probability law factorizing on a factor forest, leaf elimination gives the exact Bethe entropy identity

$$H(P_{C|R}) = \sum_x H(S_{N_C(x)}) - \sum_{v \in C} (\deg(v) - 1) H(S_v).$$

Subtract this from $\sum_v H(S_v)$ to obtain (11). Since $Q_{1/4} = Q_R \otimes \text{Ber}(1/5)^C$, the KL chain rule gives

$$D_{\text{KL}}(P \| Q_{1/4}) \geq \mathbb{E}_R D_{\text{KL}}(P_{C|R} \| \text{Ber}(1/5)^C).$$

For any law ν and product law $\otimes_v q_v$,

$$D_{\text{KL}}(\nu \| \otimes_v q_v) = \text{TC}(\nu) + \sum_v D_{\text{KL}}(\nu_v \| q_v) \geq \text{TC}(\nu),$$

which proves (12). □

3 Scalar charges and the critical pair

Lemma 3 (Scalar reduction). *The only negative charges are s_3 and s_4 :*

$$D_d = 0 \quad (d \neq 3, 4), \quad 0 < D_3 < D_4.$$

Furthermore, with $\tau := t_6$,

$$t_d \geq \tau > D_4 \quad (d \geq 6). \quad (13)$$

Proof. The finite cases $d \leq 80$ are checked by exact rational logarithmic intervals. For $d \geq 81$ put

$$L = \frac{\log(5/4)}{2 \log 2}.$$

Since $\ell_d/(2 \log 2) \leq L$,

$$s_d \geq a(d-1) - L > 0, \quad t_d \geq a - \frac{a+L}{d}.$$

The last lower bound increases with d . The exact audit checks $t_d \geq t_6$ for $6 \leq d \leq 80$ and verifies that the tail lower bound at $d = 81$ already exceeds t_6 . Thus $t_d \geq \tau = t_6 > D_4$ for every $d \geq 6$. $\square$

Put

$$\kappa = \mathcal{R}(4, 4) = \frac{\log(290/289)}{2 \log 2}. \quad (14)$$

We also need the least pair mutual information that can occur at an active factor.

Lemma 4 (Cavity odds). *Suppose an active factor has exactly two C -neighbours, each of degree 3 or 4 in A . After all other factors are absorbed into cavity messages, their joint law has unnormalised weights*

$$(2, u, v, uv), \quad \frac{1}{8} \leq u, v \leq 1.$$

Proof. For a factor of the form (10), the outgoing odds ratio toward a variable is between $1/2$ and 1 : setting that variable to 1 makes the factor identically 1 , whereas setting it to 0 can increase the corresponding partial partition function by at most a factor 2 . A C -vertex of degree d has $d-1$ other incident factors, so its cavity odds lie in $[2^{-(d-1)}, 1]$. For $d \in \{3, 4\}$ this interval is contained in $[1/8, 1]$. $\square$

Lemma 5 (Critical pair mutual information). *For the probability law with weights $(2, u, v, uv)$ and $1/8 \leq u, v \leq 1$, mutual information is minimized at $u = v = 1/8$. Hence*

$$\frac{I(X; Y)}{2 \log 2} \geq \iota, \quad (15)$$

where

$$\iota = \frac{1}{2 \log 2} \frac{1}{145} \left[128 \log \frac{290}{289} + 16 \log \frac{145}{153} + \log \frac{145}{81} \right].$$

Proof. Let $I(u, v)$ denote the mutual information. Direct differentiation gives

$$\frac{\partial I}{\partial u} = -\frac{N(u, v)}{(uv + u + v + 2)^2},$$

where

$$N(u, v) = (v+1)(v+2) \log \frac{v+1}{v+2} + v \log \frac{u+1}{u+2} + 2(v+1) \log 2.$$

For fixed v , $N(u, v)$ is increasing in u , so $N(u, v) \leq N(1, v) =: F(v)$. Differentiation gives

$$F'(v) = 1 + \log h(v), \quad h(v) = \frac{8}{3} \left(\frac{v+1}{v+2} \right)^{2v+3}.$$

Now

$$\frac{d}{dv} \log h(v) = 2 \log(1-y) + \frac{y(2-y)}{1-y}, \quad y = \frac{1}{v+2}.$$

Using the power series for $\log(1 - y)$, the coefficients of y and y^2 cancel and all remaining coefficients are positive; hence h is increasing. Therefore

$$h(v) \leq h(1) = \frac{256}{729}.$$

The exact audit verifies $1 + \log(256/729) < 0$, so $F'(v) < 0$ on $[1/8, 1]$. It also verifies $F(1/8) < 0$. Thus $N(u, v) < 0$ and $\partial I / \partial u > 0$. By symmetry I is increasing in both variables, so the minimum is attained at $(1/8, 1/8)$. There the joint probabilities are $(128, 8, 8, 1)/145$, which yields the stated value of ι . $\square$

Define

$$\zeta = 2D_4 - \kappa - \frac{\iota}{2}. \quad (16)$$

The exact audit proves $\zeta > 0$.

4 Universal local factor inequalities

A factor $x \in A$ is *clean* if $N(x) \cap R = \emptyset$ and *contaminated* otherwise. Every positive scalar charge of a C -vertex is divided equally among its d_v incident A -factors, contributing t_{d_v} on each incidence.

The following elementary tail device removes the only apparent arity cutoff. For $q \geq 1$ put

$$\rho_q = \frac{1}{2 \log 2} \log \frac{(34/33)^q + (32/33)^q}{2}. \quad (17)$$

If all residual degrees lie in $\{1, \dots, 5\}$, then $p_d \geq 1/33$, and monotonicity of (9) in each p_d gives $\mathcal{R} \geq \rho_q$.

Lemma 6 (Log-convex arity tail). *For every $q \geq 10$,*

$$\rho_q \geq (q - 1)D_4, \quad (18)$$

$$\rho_q \geq qD_4 - \zeta. \quad (19)$$

In addition, for every $q \geq 9$,

$$\rho_q + \tau \geq qD_4. \quad (20)$$

Proof. Set $U_q = (34/33)^q + (32/33)^q$. Then

$$U_{q+2}U_q - U_{q+1}^2 = \left(\frac{34 \cdot 32}{33^2}\right)^q \left(\frac{2}{33}\right)^2 \geq 0,$$

so U_q is log-convex and the increments $\rho_{q+1} - \rho_q$ are nondecreasing. The exact audit verifies

$$\rho_{10} - 9D_4 > 0, \quad \rho_{11} - \rho_{10} - D_4 > 0, \quad \rho_{10} - 10D_4 + \zeta > 0,$$

and also

$$\rho_9 + \tau - 9D_4 > 0, \quad \rho_{10} - \rho_9 - D_4 > 0.$$

The log-convexity makes the successive increments nondecreasing. These endpoint checks prove (18), (19), and (20) on their full infinite ranges. $\square$

Lemma 7 (Clean-factor inequality). *Root a clean factor component at a C -vertex. At every clean factor x , let one incident C -vertex be its parent and all other incident C -vertices its children. Then*

$$\mathcal{R}_x + \mathcal{I}_x + \sum_{v \in N_C(x)} t_{d_v} \geq \sum_{v \in \text{children}(x)} D_{d_v}, \quad (21)$$

where $\mathcal{I}_x$ is the normalized factor multi-information from Lemma 2. It suffices to use $\mathcal{I}_x = 0$ except when x has exactly two C -neighbours and both have degree 3 or 4, in which case $\mathcal{I}_x \geq \iota$.

Proof. First suppose the parent incidence has degree at most 5. Every degree- ≥ 6 child contributes at least $\tau > D_4$ and may therefore be paired with and pay one dangerous child. Remove such paid pairs. Removing variables can only decrease the collision term, so this operation is conservative. If no dangerous child remains, the claim is immediate. Otherwise all residual degrees lie in $\{1, \dots, 5\}$ and the parent remains present. If the residual arity is $q \geq 10$, the child deficit is at most $(q-1)D_4$, and (18) closes the inequality. The cases $2 \leq q \leq 9$ are checked exactly; $q \leq 1$ is trivial.

Now suppose the parent itself has degree at least 6. Keep that parent and retain its full incidence credit, which is at least $\tau = t_6$. Strip only degree- ≥ 6 children, pairing each with a dangerous child as above. What remains is a multiset of low-degree children in $\{1, \dots, 5\}$. If their number is $q \geq 9$, then their total deficit is at most qD_4 and (20) gives

$$\mathcal{R}_x + \tau \geq qD_4.$$

The cases $0 \leq q \leq 8$ are checked exactly. Their smallest margin occurs for two degree-4 children and is strictly positive.

In the low-parent finite table, the unique critical configuration is the clean pair $(4, 4)$, where Lemma 5 supplies the factor multi-information and

$$\kappa + \iota - D_4 > 0.$$

This completes all arities and all parent degrees. $\square$

Lemma 8 (Contaminated-factor inequality). *Let a contaminated factor x have $k_R = |N(x) \cap R| \geq 1$. After its collision contribution and all incidence credits t_d are spent, the total unpaid dangerous C -deficit at that factor is at most*

$$\zeta k_R. \tag{22}$$

When $k_R = 1$ and the residual C -configuration is the dangerous pair $(4, 4)$, half of the pair mutual-information bound ι is also used.

Proof. The same stripping argument removes all degree- ≥ 6 neighbours unless no dangerous deficit remains. Suppose the residual degrees lie in $\{1, \dots, 5\}$ and let the residual arity be q .

If $q \geq 10$, the total deficit is at most qD_4 and Lemma 6 gives

$$\sum D_{d_v} - \mathcal{R}_x \leq \zeta \leq \zeta k_R.$$

Thus $q \leq 9$ remains.

For $k_R = 1$, the exact finite audit checks all residual multisets. In the unique critical case $(4, 4)$, the sole R -neighbour is unoccupied with probability at least $1/2$: the map $I \mapsto I \setminus \{r\}$ injects independent sets containing r into those avoiding r . Hence the factor is active with probability at least $1/2$, and Lemma 5 supplies $\iota/2$. The residual is then exactly

$$2D_4 - \kappa - \frac{\iota}{2} = \zeta.$$

For $k_R \geq 2$, no mutual-information contribution is needed. It is enough to audit $k_R = 2$, since the right side of (22) increases with k_R while the remaining local budget is independent of k_R . The exact audit verifies all $q \leq 9$ cases for $k_R = 2$. $\square$

5 The R -core budget

For $v \in R$ put $k_v = \deg_K(v) \geq 1$. Then

$$\sum_{v \in R} k_v = 2e.$$

Lemma 9 (*R*-core domination). *For every integer $d \geq 1$ and every integer $k \geq 1$,*

$$s_d + \frac{19}{1000}k \geq \zeta d. \quad (23)$$

Proof. The left side increases with k , so take $k = 1$. The exact audit checks $1 \leq d \leq 80$. For $d \geq 81$, with $L = \log(5/4)/(2 \log 2)$,

$$s_d + \frac{19}{1000} - \zeta d \geq (a - \zeta)d - a - L + \frac{19}{1000}.$$

The audit verifies $a > \zeta$ and positivity at $d = 81$, so the lower bound is positive for every larger d . $\square$

Summing (23) over $v \in R$ and splitting the edge term equally between the endpoints of every edge of K gives

$$\sum_{v \in R} s_{d_v} + ae = \sum_{v \in R} \left(s_{d_v} + \frac{19}{1000}k_v \right) \geq \zeta \sum_{v \in R} d_v. \quad (24)$$

6 Global bookkeeping

The final point is to show that the local inequalities above can be summed without double spending.

Lemma 10 (Global assignment). *Assume $R \neq \emptyset$ and fix a root $r \in R$ of the original tree T . Delete from the bipartite A – C subgraph every contaminated factor. Each remaining clean component is a tree. Root each such component at its unique C -vertex closest to r in T .*

Then:

1. *every nonroot dangerous C -vertex has a unique parent clean factor, and its deficit is charged there;*
2. *every root dangerous C -vertex has a unique parent contaminated factor in T , and its deficit is charged there;*
3. *every incidence credit t_{d_v} is used exactly once, at its incident A -factor;*
4. *every collision contribution $\mathcal{R}_x$ and every Bethe factor contribution $\mathcal{I}_x$ is used at most once;*
5. *the sum of k_R over all contaminated factors is exactly*

$$\sum_{x \text{ contaminated}} k_R(x) = \sum_{v \in R} d_v.$$

Consequently, after Lemmas 7 and 8 are summed, the total unpaid C -deficit is at most

$$\zeta \sum_{v \in R} d_v. \quad (25)$$

Proof. Because T is a tree, every connected clean component has a unique vertex closest to r and a unique edge on the path from that component toward r . That boundary edge joins a C -vertex to a contaminated A -factor; otherwise the factor would belong to the clean component. Orient every clean component away from this root C -vertex. Hence every nonroot C -vertex has exactly one parent clean factor and every root C -vertex has exactly one parent contaminated factor.

The scalar identity $s_d = dt_d - D_d$ assigns one copy of t_d to each of the d A -incidences of a C -vertex; these incidences are disjoint, proving item 3. The bounds (8) and (11) are already sums indexed by A -factors, so each such factor budget is used only at that factor, proving item 4. Lemma 7 pays every nonroot deficit. At a contaminated factor we ask Lemma 8 to pay only those adjacent C -deficits assigned to it; its inequality was proved against the larger demand consisting of all adjacent dangerous C -deficits, so it remains valid.

Finally, an edge from R to A is incident with a contaminated factor, and every incidence counted by $k_R(x)$ is exactly one such R - A edge. Counting the same edges from R gives item 5. Summing (22) proves (25). $\square$

7 Proof of Theorem 1

If T is a single vertex, then $\mu(T)/\alpha(T) = 1/2 > r_0$. Assume henceforth that $B \neq \emptyset$.

Apply (4) and decompose each C -charge by (5).

If $R = \emptyset$, the clean factor graph is the whole tree. Root it at any C -vertex. Lemma 7 pays every dangerous C -deficit except possibly the root deficit, which is at most D_4 . The constant term $a = 19/500$ satisfies $a > D_4$ by the exact audit. All remaining terms are nonnegative, so $\mu(T) - r_0\alpha \geq 0$.

Now suppose $R \neq \emptyset$. By Lemma 10, the clean and contaminated factor budgets pay all C -deficits except for a residual at most (25). The scalar charges on R and the entire edge term ae have not yet been spent. Lemma 9, summed as in (24), pays (25) in full. The constant a in (4) remains nonnegative and unused. Thus

$$\mu(T) - \frac{231}{500}\alpha(T) \geq 0.$$

This proves Theorem 1. $\square$

8 Exact audit

The accompanying script `verify_second_connectivity_0462_QED.py` uses `fractions.Fraction` only. Every logarithm is enclosed with the atanh series

$$\log x = 2 \sum_{j=0}^{N-1} \frac{y^{2j+1}}{2j+1} + R_N, \quad y = \frac{x-1}{x+1},$$

with the rational remainder bound

$$|R_N| \leq \frac{2|y|^{2N+1}}{(2N+1)(1-y^2)}.$$

It checks exactly:

- the scalar signs through $d = 80$ and the analytic scalar tail from $d = 81$ onward;
- $t_d \geq D_4$ for all $d \geq 6$;
- every reduced clean factor with low-degree parent through arity 9;
- every reduced clean factor with degree- ≥ 6 parent through eight low-degree children;
- both log-convex clean arity tails, including the high-parent case;
- every reduced contaminated factor of arity at most 9 for $k_R = 1$ and for the worst case $k_R = 2$ among all $k_R \geq 2$;

- the contaminated arity tail from $q = 10$ onward;
- the two endpoint logarithmic inequalities used in the mutual-information monotonicity proof;
- the R -core inequality through $d = 80$ and its analytic tail.

The executed audit terminates with PASS UNIVERSAL QED AUDIT.

Remark 11. *The proof deliberately targets the rational value $231/500$. It does not claim optimality of the method. The clean $(4, 4)$ factor retains a small positive margin, so any further improvement must quantify additional structure rather than merely re-use the same local pair inequality.*